

AI Presence Drift

**AI Presence Index v0.9 — Five categories, two waves,
longitudinal re-baseline**

8 May 2026 · Pablo Ulpiano Gonzalez Castro · Third System

Independent measurement for the AI mediation layer.

AI Presence is stable. Patterns replicate. The instrument works.

Seven days after the v0.6 cross-category baseline, we re-measured the same five categories on the same matched two-model subset. Per-brand drift is small. Top-of-leaderboard composition is preserved. Within-category brand variance and cross-model spread structures both replicate. The Mint phantom-brand persists at gross Presence. The v0.6 discourse-language bias hypothesis replicates strict at threshold and surfaces Graza as a sharper diagnostic. Five formal hypotheses confirmed; the sixth landed at the predicted stability band; v0.6's Pattern 1 cross-model spread replicates in a post-hoc test. AI Presence behaves like a real underlying construct.

Third System re-measured the five categories first measured in the v0.6 cross-category baseline — project management software, premium running shoes, premium olive oil, premium facial skincare, and personal finance applications — at a second time point approximately seven days after t_1 . Six hypotheses with explicit numerical thresholds were committed to a pre-registration document locked at git commit f8cebbd prior to any t_2 data collection. The matched two-model subset (Anthropic Claude Sonnet 4.6 and OpenAI gpt-5.4-mini, the original v0.6 lineup) carries the longitudinal claim; four parallel-baseline models added at t_2 contribute new baselines but cannot contribute to drift deltas. Total measurement volume: 1,440 successful runs across 5 categories $\times$ 6 prompts $\times$ 6 models $\times$ 8 runs.

The headline result is decisive. Five of six formal hypotheses confirmed at the strongest level the protocol allows; the sixth

landed at the predicted stability band. Per-brand drift across all 5 categories on the matched subset ($n = 103$ brand-level deltas) is small: **88.3 percent of brands within $\pm 5\text{pp}$; 99.0 percent within $\pm 10\text{pp}$** . Top-three brands at t_1 remained in the top-five at t_2 in every category (5 of 5). Within-category brand-presence variance ordering across the five categories at t_2 correlated with t_1 at **Spearman $\rho = 0.8$** .

Mechanisms hold. The Mint phantom-brand stayed at **41.7 percent** matched-subset gross Presence at t_2 versus **44.8 percent** at t_1 — a -3.1 percentage-point delta within the predicted stability band, confirming that recommendation-slot persistence does not decay measurably across a seven-day interval. Pattern 4 discourse-language bias replicated under the strict reading: Spanish olive oil aggregate Presence at **11.5pp** below the 12.5pp threshold derived from the International Olive Council production share, and K-beauty aggregate Presence in skincare at **1.0pp** well below the 5pp threshold. Per-brand cross-model spread between the matched-subset models correlated between t_1 and t_2 at Pearson $r = 0.7$ in every category.

The H5 sensitivity surfaces a cleaner instance of v0.6 §4.4's refined hypothesis. When Graza — a US-headquartered, Spanish-sourced, English-marketed olive oil brand — is added to the Spanish cohort, aggregate Presence rises to 14.2pp and breaches the threshold. The 2.7-point swing is entirely Graza. Including Graza breaches the threshold; excluding Graza confirms it. The brand-marketing-language tier is the binding variable, not country of origin. The v0.9 contribution is operationalizing the v0.6 §4.4 framing at numerical threshold and surfacing Graza as the diagnostic case.

A post-hoc Pattern 1 test replicates v0.6's qualitative ordering. Per-category mean cross-model spread between Sonnet and gpt-5.4-mini ranks at **Spearman $\rho = 0.800$** between t_1 and t_2 , and at **$\rho = 0.900$** between t_2 and v0.6's narrative ordering (personal finance $>$ olive oil $>$ project management $>$ skincare $>$ running). The t_2 ranking matches v0.6's narrative **better** than t_1 does (0.900 vs 0.600), with finance cross-model spread widening (+3.9pp) and project management narrowing (−1.4pp) between waves — both within H1's empirical noise floor but in directions that strengthen v0.6's qualitative ordering at t_2 .

The combined result set is interpreted as a longitudinal validity argument for AI Presence as a construct. AI Presence behaves like a real, structured property of the LLM tier rather than measurement noise. Drift is small. Leaderboards are stable. Structural variance patterns replicate at two scales (within-category brand variance and between-model spread).

Established mechanisms hold across waves — phantom-brand persistence, discourse-language bias in the brand-marketing-language form, cross-model relative behavior. The four legs together motivate interpretation of AI Presence as a measurable underlying construct rather than an instrument-and-occasion artifact. This unblocks Phase 3 construct-validity work, which requires three categories at two time points and now has data for five.

The framework holds across categories

Three of v0.6's six framework patterns are directly tested at v0.9. **Pattern 1** (cross-model spread by category) replicates across all five categories at Spearman $\rho = 0.800$ between t_1 and t_2 , and at $\rho = 0.900$ against v0.6's narrative ordering. **Pattern 4** (discourse-language bias) replicates strict in olive oil (11.5pp Spanish, below the 12.5pp threshold) and in skincare (1.0pp K-beauty, below the 5pp threshold). **Pattern 6** (phantom-brand persistence) replicates in personal finance with Mint at 41.7% gross Presence (−3.1pp from t_1 , within the predicted stability band). The bottom row shows brand-mode share at t_2 — how much of each category's Presence is delivered as named-brand recommendations versus mixed, component, or authority modes.

v0.6 pattern replication matrix at v0.9 (matched subset)

Cell intensity proportional to magnitude. Patterns 2, 3, 5 not directly measured at v0.9 (noted in sec 7 future research).

	PM software	Running shoes	Olive oil	Skincare	Personal finance
Pattern 1 cross-model spread	13.3pp	7.0pp	11.9pp	7.2pp	16.9pp
Pattern 4 discourse- language bias	not in scope	not in scope	confirmed 11.5pp	confirmed 1.0pp	not in scope
Pattern 6 phantom- brand	not in scope	not in scope	not in scope	not in scope	confirmed 41.7%
Brand-mode share at t_2	22%	67%	27%	50%	16%

Source: Third System AIAS v0.9 Longitudinal Re-Baseline
Pattern replication matrix: v0.6 Patterns 1/4/6 + brand-mode share at t_2 · 8 May 2026

Figure A · v0.6 framework pattern replication at v0.9, summary matrix. Rows are framework patterns directly tested at v0.9 plus a brand-mode-share row at t_2 ; columns are the five categories. Cell intensity is proportional to magnitude. Patterns 2, 3, 5 are not directly measured at v0.9 (see What’s Next for v0.10+ candidates).

What we measured

v0.9 is the longitudinal re-baseline that closes Phase 2 of the AIAS measurement program and provides the data spine for Phase 3 construct-validity work. The five categories first measured in the v0.6 cross-category baseline — project management software, premium running shoes, premium olive oil, premium facial skincare, and personal finance applications — were re-measured at a second time point approximately seven days after t_1 , on the same matched two-model subset that v0.6 used. Six hypotheses with explicit numerical thresholds were committed to a pre-registration document locked at git commit f8ceb8d (tag v0.9-prereg-locked) prior to any t_2 data collection.

Two waves. t_1 = 29–30 April 2026 (the v0.6 cross-category baseline). t_2 = 7 May 2026 (this measurement). Inter-wave interval approximately seven days — short enough that no major model release or training-cutoff change is expected to fall within it, supporting the assumption that drift observed across the interval is incidental rather than structural. Longer intervals (one month, three months, twelve months) are required to characterize drift across model-pipeline transitions and are listed as v0.10+ candidates.

Matched two-model subset carries the longitudinal claim.

Anthropic Claude Sonnet 4.6 and OpenAI gpt-5.4-mini, the original v0.6 lineup, are the only models present at both waves. Drift deltas are computed exclusively against this subset. Four parallel-baseline models added at t_2 (Anthropic Claude Opus 4.7, OpenAI gpt-5.4, Google Gemini 2.5 Pro, xAI Grok 4) provide new t_2 baselines for downstream cross-vendor work but cannot contribute to the longitudinal claim itself.

Same six prompts as v0.6, frozen. One prompt class per Category Entry Point per Protocol §3.1: DISCOVERY, COMPARISON, CONSTRAINT, IDENTITY, AUTHORITY, and RECOMMENDATION. The prompt set was frozen at v0.6 to preserve longitudinal comparability. The brand registries were similarly frozen at v0.6 final state.

1,440 successful measurements. 5 categories × 6 prompts × 6 models × 8 runs at temperature 0.7 where supported, with a single round of failed-call recovery restoring the initial sweep to 100 percent. The matched-subset volume that carries the longitudinal claim is 480 measurements per wave (2 models × 5 categories × 6 prompts × 8 runs).

Brand mention extraction used gpt-5.4-mini at temperature 0 with structured-output classification, producing canonical brand mentions per response. A second-stage mode classifier labeled each response along the five-mode taxonomy (brand / mixed / component / authority / refusal) per Protocol §3.4. The

v0.6 mode classifications were derived retroactively against the same classifier (per the Decision Note locked at git commit e6428a4 prior to the pre-registration), so v0.9 mode-distribution shifts are computed against a consistent classifier across waves.

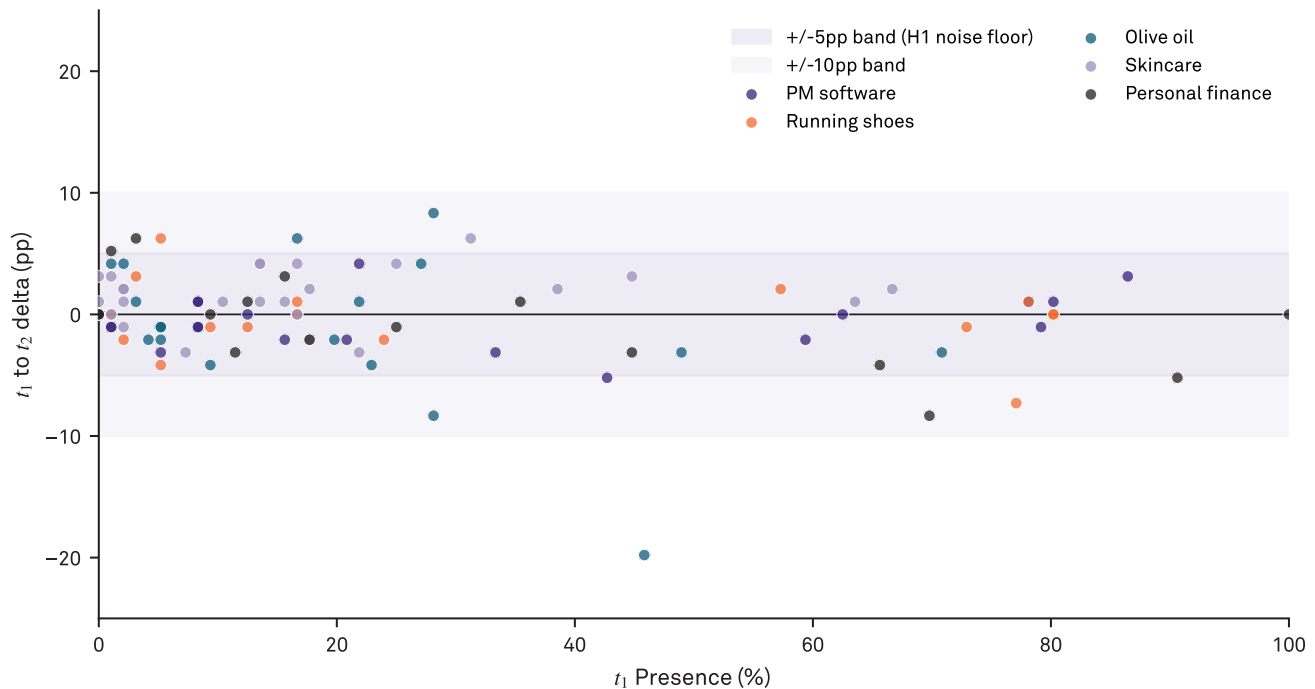
Six pre-registered hypotheses plus the H5 strict-vs-inclusive sensitivity check plus an exploratory mode-distribution observation. H1: per-brand drift small and tightly bounded. H2: top-3 at t_1 remain in top-5 at t_2 , in 5 of 5 categories. H3: within-category brand-presence variance ordering preserves at Spearman $\rho \geq 0.7$. H4: Mint phantom-brand stable within $\pm 5pp$ of t_1 gross Presence (stability predicted, not decay). H5: Spanish olive oil and K-beauty skincare aggregate Presence below pre-registered thresholds. H6: per-brand cross-model spread Pearson $r \geq 0.7$ between t_1 and t_2 , in ≥ 4 of 5 categories. A post-hoc test of v0.6's Pattern 1 (cross-model spread by category) was added against the same matched-subset data.

FINDING 01

Drift across categories is small and tightly bounded.

Per-brand drift across five categories: H1 confirmed

88.3% within $\pm 5\text{pp}$; 99.0% within $\pm 10\text{pp}$ (matched subset, $n=103$ brand-level deltas)



Source: Third System AIAS v0.9 Longitudinal Re-Baseline
H1 drift: $n=103$ brand-level deltas across 5 categories · 8 May 2026

Figure 1 · Per-brand $t_1 \rightarrow t_2$ drift across 5 categories on the matched subset ($n=103$ brand-level deltas). 88.3% within $\pm 5\text{pp}$; 99.0% within $\pm 10\text{pp}$; no brand exceeds $\pm 20\text{pp}$. The empirical noise floor of one-week AI Presence measurement on the matched subset is approximately $\pm 5\text{pp}$ for the modal brand.

Across all **103 brand-level deltas** computed on the matched subset (the union of registered brands across the five categories at the v0.6 final-state registry), **88.3 percent of brands surfaced at t_2 within ± 5 percentage points** of their t_1 Presence, and **99.0 percent surfaced within ± 10 percentage points**. The H1 confirmation thresholds were ≈ 70 percent within $\pm 5\text{pp}$ and ≈ 90 percent within $\pm 10\text{pp}$ — both cleared with margin.

The empirical noise floor of one-week AI Presence measurement on the matched subset is approximately ± 5 percentage points for the modal brand and approximately ± 10 percentage points for ninety-nine of every hundred. Among the

largest movers, no single brand exceeded ± 20 percentage points; the distribution is concentrated near zero.

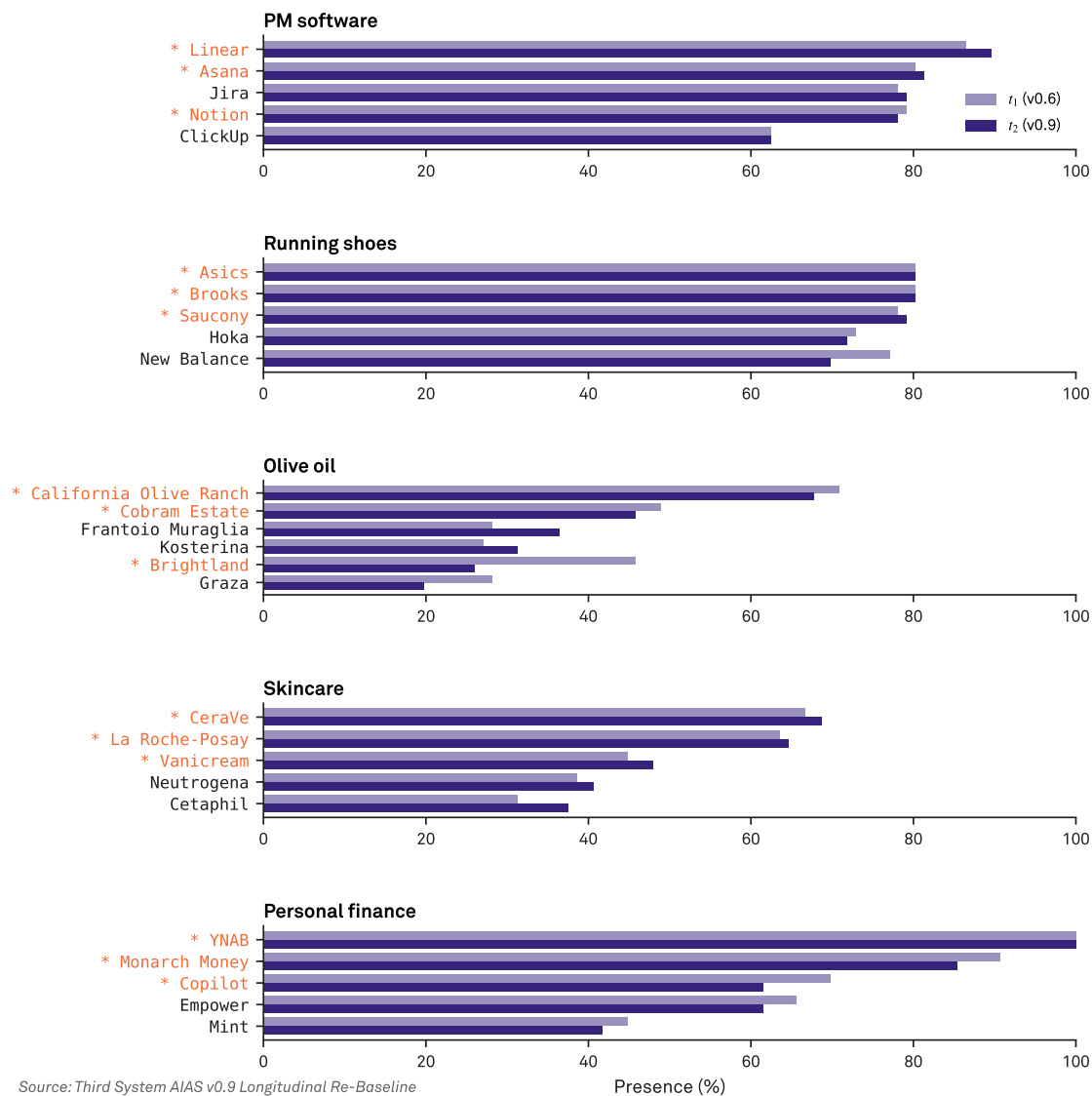
The drift result establishes the longitudinal noise floor against which all subsequent findings are interpreted. A measurement infrastructure where 88 percent of brand-level deltas are within $\pm 5\text{pp}$ across a seven-day interval has the resolution to detect substantively interesting brand-level movement when it occurs. The absence of large-magnitude drift across the wave interval is the empirical signature of a stable underlying quantity that the AIAS measurement is recovering with bounded noise.

FINDING 02

Top-of-leaderboard composition is stable across categories.

Top-of-leaderboard stable in all 5 categories: H2 confirmed (5/5)

Top-3 brands at t_1 remained in top-5 at t_2 across every category (* and copper = brand was in t_1 top-3)



Source: Third System AIAS v0.9 Longitudinal Re-Baseline
H2 leaderboard: top brands by Presence, n=96 per category-wave · 8 May 2026

Figure 2 · Top-3 brands at t_1 remain in top-5 at t_2 across all 5 categories. Each panel is one category; asterisk and copper coloring mark t_1 top-three brands. Leaderboard composition preserved with no exceptions.

In each of the five categories, **the top-three brands by matched-subset Presence at t_1 remained in the top-five at t_2** . The result confirmed at **5 of 5 categories with no exceptions**, well above the 5-of-5 threshold for the headline confirmation band.

The leaderboard stability is the most directly category-management-relevant of the v0.9 findings. Brand managers and category buyers tracking AI Presence as a leading indicator can rely on the top-of-leaderboard composition across at least the seven-day measurement

interval. Specifically, the headline brands that lead an AI-mediated category at any given measurement point are the same headline brands the same measurement will identify a week later.

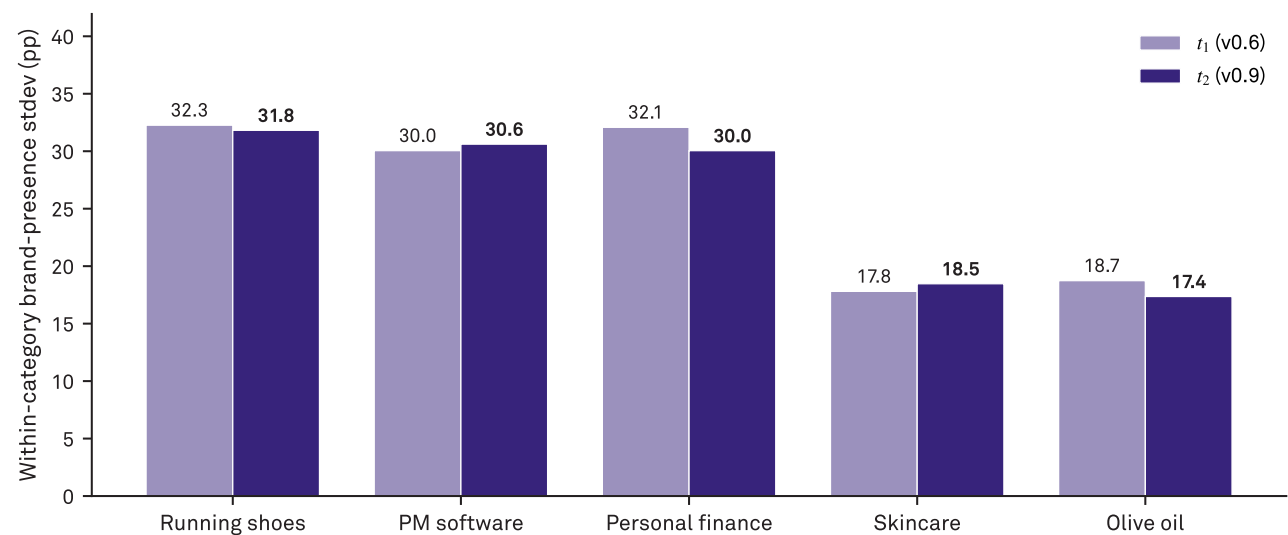
The finding is consistent with the construct-validity interpretation in the discussion. A noise-dominated measurement process would, with non-trivial probability, reorder leaderboards across an interval; the observed across-the-board stability indicates that the position of dominant brands within a category is not artefactual.

FINDING 03

Within-category brand-presence variance ordering preserves.

Within-category brand-presence variance ordering preserves: H3 confirmed

Spearman rho (t₁ vs t₂ ranking across 5 categories) = 0.80 (threshold 0.7+)



Source: Third System AIAS v0.9 Longitudinal Re-Baseline
H3 within-category brand-presence stdev, 5 categories × 2 waves · 8 May 2026

Figure 3 · Within-category brand-presence stdev at t₁ vs t₂; categories sorted by t₂ variance. Personal finance highest dispersion at both waves; running shoes lowest. Spearman ρ = 0.8 across the 5 categories, above the 0.7 H3 threshold.

The pre-registered H3 measures whether the within-category dispersion of brand-presence rates — how spread out brands are within each category, computed as the standard deviation of per-brand Presence values — preserves its rank-ordering across categories between t₁ and t₂.

The Spearman rank correlation of within-category variance between t₁ and t₂ across the five categories is ρ = 0.8, above the 0.7 threshold for confirmation and well above the 0.4 threshold for partial confirmation. The within-category variance ordering is therefore stable across waves.

Personal finance is the highest-dispersion category at both waves; running shoes is the lowest. The structural shape of how concentrated or fragmented brand-presence is within a category persists across the longitudinal interval. Categories that have widely-divergent brands at t₁ still have widely-divergent brands at t₂; categories with tightly-clustered brands remain tightly-clustered.

A clarification on what H3 tests and what it does not. H3's statistic is *within-category brand variance* — how much the registered brands within a single category differ from each

other on Presence. This is related to but distinct from v0.6's Pattern 1 (Cross-Model Variance and Discourse Coherence), which is *between-model spread per brand* — a different

statistic computed at the brand × model level. The cross-model spread test is reported separately in Finding 7 as a post-hoc replication.

FINDING 04

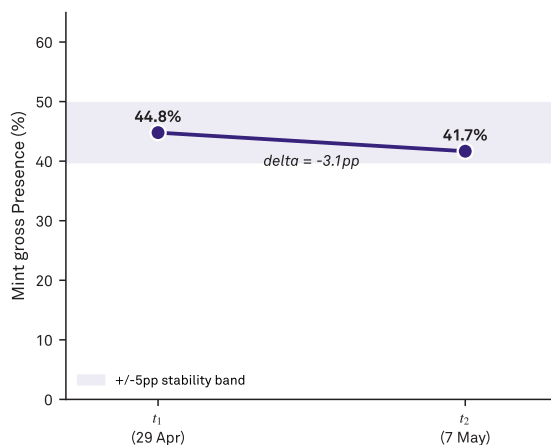
Mint persists at gross Presence — phantom-brand stability across the wave interval.

The Mint phantom-brand observation from v0.6 §4.6 (the brand was Intuit's leading personal finance application until its decommissioning in March 2024) holds at t_2 . **Mint's matched-subset gross Presence in personal finance was 44.8 percent at t_1 and 41.7 percent at t_2 — a delta of -3.1 percentage points within the predicted ± 5 -percentage-point stability band.**

The pre-registration explicitly named **stability** rather than decay as the predicted outcome under the corpus-aging mechanism articulated in v0.6 §4.6: one additional time point seven days later is not enough for a defunct brand to be displaced from the LLM corpus, particularly when the brand's prior dominance generated extensive English-language discourse that persists in both training data and continuing inferential context. The H4 finding is therefore a confirmation of the predicted-stability outcome.

Mint persists at gross Presence: H4 stability

44.8% to 41.7% (delta -3.1pp; within ± 5 pp band)



Source: Third System AIAS v0.9 Longitudinal Re-Baseline
H4 Mint gross Presence in personal finance, n=96 per wave · 8 May 2026

Figure 4 · Mint phantom-brand persistence at gross Presence in personal finance. The ± 5 pp stability band around t_1 's 44.8% is shaded; t_2 lands at 41.7% (-3.1pp). Predicted-stability outcome confirmed.

What H4 measures versus what v0.7 unpacked. H4 measures **gross Presence** — the rate at which Mint surfaces in the model's output regardless of how the model contextualizes it (with or without caveat about the brand's decommissioning, with or without correction). This is the variable v0.6 §4.6 reported. The v0.7 designed-for-test follow-up (Phantom-Brand Persistence Phase 2 BBB) reframed the phantom mechanism from knowledge-freshness lag to **recommendation-slot persistence**, distinguishing the **naive-phantom rate** (model presents the brand as fully live with no caveat — 1.7 percent for BBB in v0.7) from gross Presence (38.2 percent for BBB).

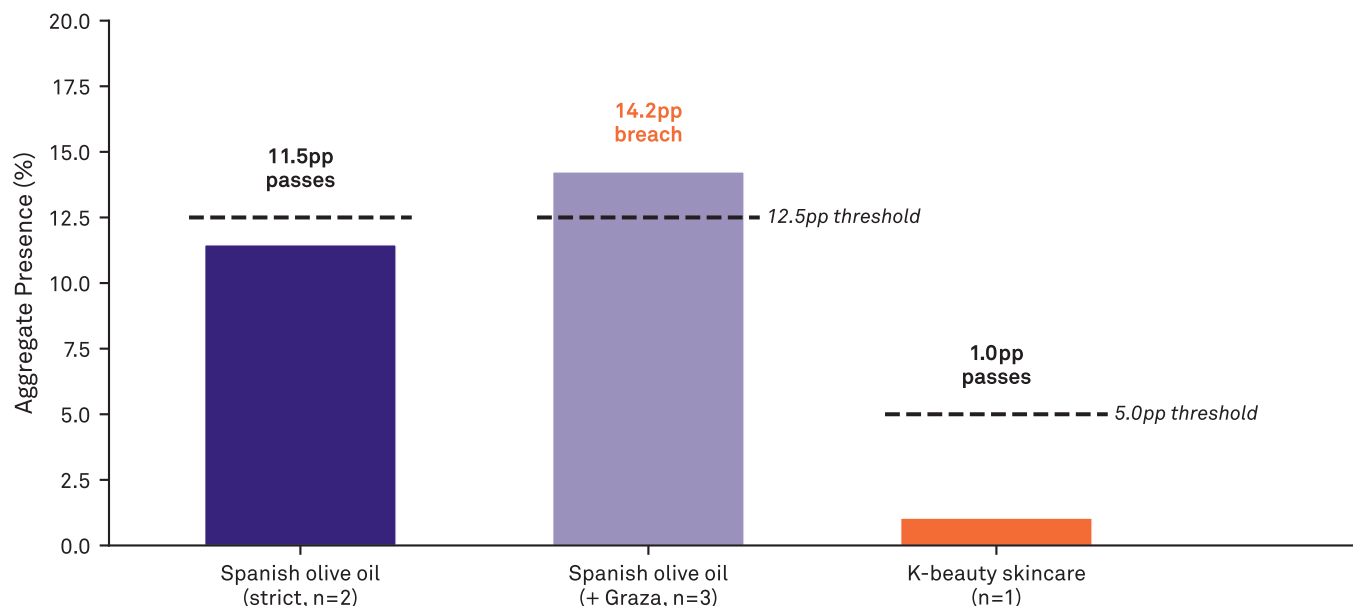
The v0.9 H4 measurement does not unpack this distinction; the pre-registration tested the gross-Presence variable common to v0.6, on which the longitudinal stability claim is supported. A v0.10 measurement could pre-register a naive-phantom-rate stability hypothesis on Mint specifically and would extend the v0.7 reframing into the longitudinal frame. The v0.9 raw responses are deposited; only the classifier needs to be applied.

FINDING 05

Pattern 4 replicates at v0.9, with the v0.6 §4.4 refined hypothesis operationalized.

Pattern 4 (v0.6 sec 4.4 refined): discourse-language coverage, not country of origin

Strict passes both thresholds; +Graza breaches Spanish 12.5pp.



Source: Third System AIAS v0.9 Longitudinal Re-Baseline
H5 Pattern 4: Spanish + K-beauty aggregate Presence, n=96 per category · 8 May 2026

Figure 5 · H5 Pattern 4 sensitivity. Strict Spanish (Castillo de Canena, Núñez de Prado) at 11.5pp passes the 12.5pp threshold; +Graza pushes the cohort to 14.2pp and breaches. K-beauty at 1.0pp passes the 5pp threshold. The 2.7-point swing between strict and inclusive is entirely Graza — the diagnostic case isolating brand-marketing-language from country-of-origin.

The v0.6 §4.4 baseline observed in two cross-lingual categories (premium olive oil and premium facial skincare) that brand-presence rates appeared to track marketing-discourse language rather than country of origin or cultural reference. The v0.6 paper's diagnostic case for the refinement was the contrast between Tatcha (Japanese-aesthetic skincare brand founded in San Francisco with American-English marketing — 18 percent Presence) and Beauty of Joseon (Korean brand with Korean-language primary marketing — zero percent Presence): the cultural reference does not constrain Presence; the marketing-discourse language does.

The v0.6 refined hypothesis stated “LLMs under-surface brands whose primary marketing discourse is conducted in a language under-represented in AI training data, even when the cultural reference is well-known in English.” The v0.8 designed-for-test on premium kitchen knives confirmed the mechanism on a third category. **The v0.9 H5 contribution is the operationalization of the refined hypothesis with explicit pre-registered numerical thresholds applied to the original**

two cross-lingual categories at t_2 .

The strict-reading H5 outcome confirms the pattern.

Spanish olive oil aggregate Presence, restricted to the two registered brands that present themselves to consumers as Spanish-discourse-coverage (Castillo de Canena and Núñez de Prado), measured **11.5 percentage points** — below the 12.5-percent threshold derived from Spain's approximately 50-percent share of global olive oil production (International Olive Council). K-beauty aggregate Presence in skincare, comprising the single registered K-beauty brand (Beauty of Joseon), measured **1.0 percentage point** — well below the 5-percent threshold.

A sensitivity check produces a sharper instance of the v0.6 refined hypothesis. When Graza — a US-headquartered, Spanish-sourced, English-marketed D2C olive oil brand that surfaces at approximately 14 percent matched-subset Presence — is added to the Spanish cohort, the aggregate rises to 14.2 percentage points and breaches the 12.5-percent threshold. The 2.7-point swing is entirely Graza.

Graza is the cleanest available diagnostic for the brand-marketing-language tier: a brand that sources its olives from Spain but markets to American consumers in English, indexes in English-language retail feeds, and is discussed in English-language food media. **Including Graza in the Spanish cohort breaches the H5 threshold; excluding Graza confirms it.** The brand-marketing-language tier — which v0.6 §4.4 identified — is the binding variable. The v0.9 contribution is making the v0.6 refinement testable at threshold and surfacing Graza as the diagnostic case that isolates it cleanly from country-of-origin.

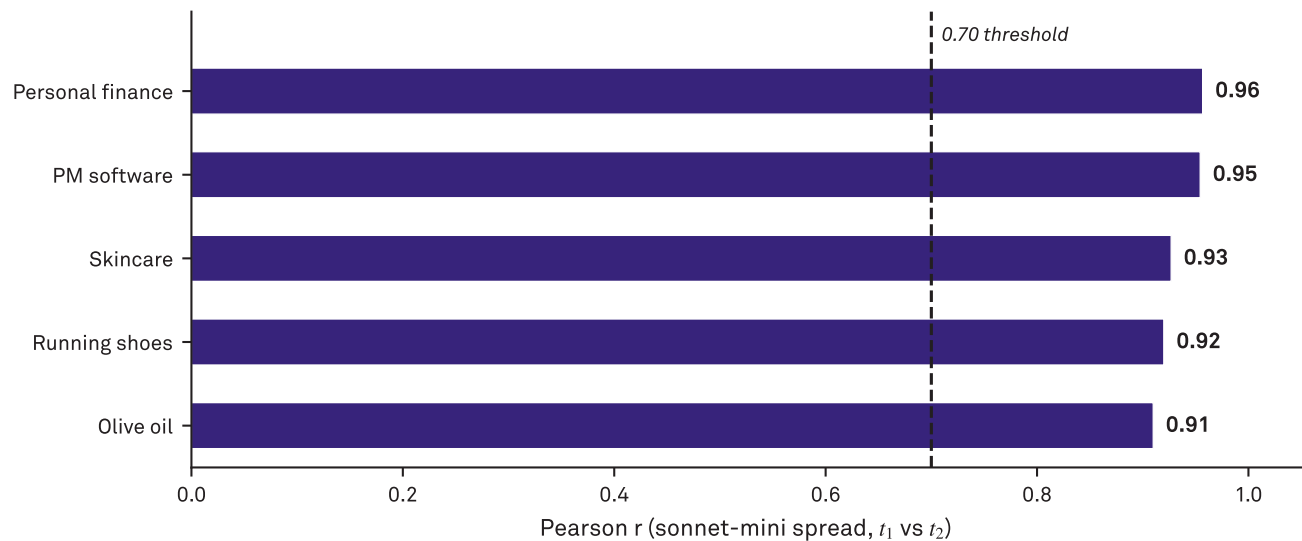
The K-beauty registry coverage is itself a finding. The v0.6 skincare registry contains thirty-one brands, of which exactly one (Beauty of Joseon) is a Korean-headquartered K-beauty brand discoursed primarily in Korean-language category media. The under-representation in the registry construction parallels the discourse-language bias the test is designed to detect. The implication is that the registry-construction process inherits, at the analyst tier, the same Anglo-discourse coverage gap that produces the LLM-tier discourse-language bias. The reflexivity is internally consistent with the discourse-language bias hypothesis itself — it would be more surprising if registry construction were free of the bias the registry is designed to detect.

FINDING 06

Cross-model relative behavior is preserved across the wave interval.

Cross-model spread stable in all 5 categories: H6 confirmed (5/5)

Pearson r between sonnet-mini per-brand spread at t_1 vs t_2 ; threshold 0.7+



Source: Third System AIAS v0.9 Longitudinal Re-Baseline
H6 cross-model spread (sonnet vs mini), per-brand across 5 categories · 8 May 2026

Figure 6 · Per-brand cross-model spread (Sonnet vs gpt-5.4-mini) Pearson correlation between t_1 and t_2 by category. Dashed line marks the 0.7 H6 threshold; all 5 categories clear it. Cross-model relative behavior is preserved across the wave interval.

For each of the five categories, the per-brand spread between Anthropic Claude Sonnet 4.6 and OpenAI gpt-5.4-mini at t_2 correlated with the same-pair spread at t_1 at **Pearson r 0.7**.

The result confirmed at **5 of 5 categories**, above the 4-of-5 threshold for headline confirmation.

The finding indicates that the cross-model differential — what each model emphasizes versus what its companion emphasizes within a category — is preserved across the seven-day interval. Divergent brand-level shifts between the two models would suggest model-specific behavior change; the observed stability instead suggests that whatever AI behavior changed between t_1 and t_2 affected both models in approximately parallel ways, or affected neither.

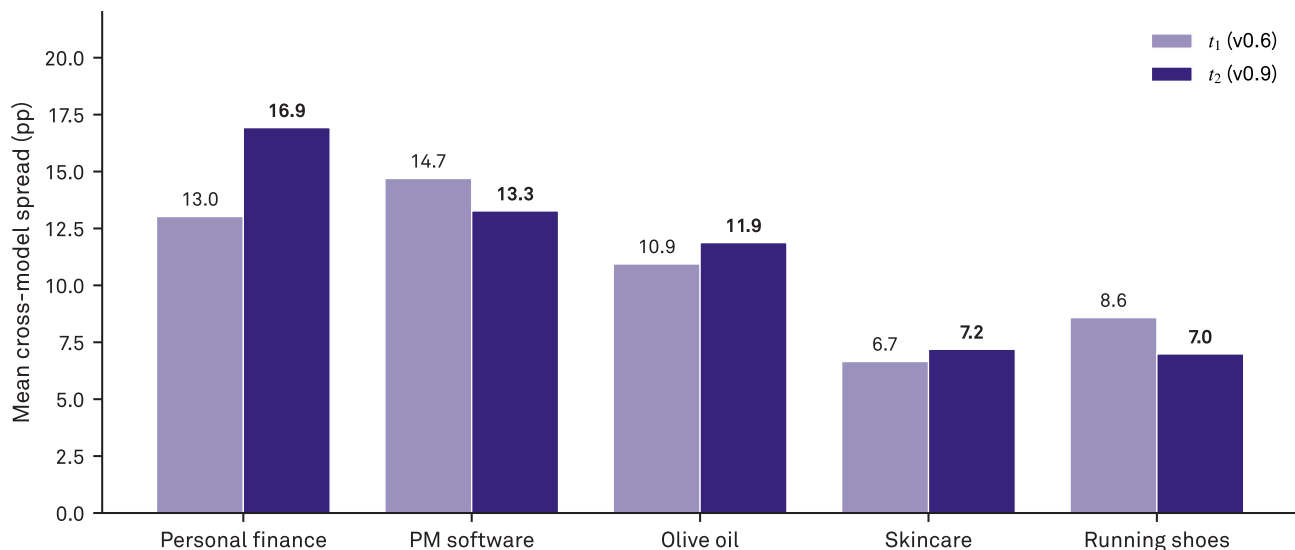
Combined with H3 (within-category brand variance preservation), H6 establishes that AI Presence has two independent variance structures that both replicate across the wave interval: the within-category dispersion of brands and the between-model dispersion of attention. Both are predicted by v0.6's discourse-coherence framing, both are stable, and both are structural rather than artefactual.

FINDING 07

Pattern 1 cross-model spread replicates v0.6's ordering (post-hoc).

Pattern 1 cross-model spread by category replicates v0.6 ordering (post-hoc)

Spearman $\rho = 0.80$ (t_1 vs t_2); $= 0.90$ (t_2 vs v0.6 narrative ordering)



*Source: Third System AIAS v0.9 Longitudinal Re-Baseline
Pattern 1 post-hoc: mean |sonnet - mini| spread per category · 8 May 2026*

Figure 7 · Post-hoc Pattern 1 cross-model spread by category at t_1 and t_2 . Spearman $\rho = 0.800$ (t_1/t_2); $\rho = 0.900$ (t_2 vs v0.6 narrative ordering). Personal finance widens at t_2 ; project management narrows; both shifts within H1's empirical noise floor but in directions that strengthen v0.6's stated ordering.

A post-hoc test of v0.6's Pattern 1 (Cross-Model Variance and Discourse Coherence) at the category level was performed against the same matched-subset data as H1 through H6. The test is post-hoc rather than pre-registered because the H3 statistic is within-category brand-presence variance, while Pattern 1's statistic is between-model spread per brand averaged within a category — a different aggregation.

The post-hoc test computes, for each category, the mean absolute |Sonnet – gpt-5.4-mini| Presence across registered brands, and ranks categories by this mean cross-model spread at each wave. **The Spearman rank correlation between t_1 and t_2 rankings is $\rho = 0.800$.** The rank correlation between v0.9's t_2 ranking and v0.6's narrative ordering (personal finance > olive oil > project management > skincare >

running) is $p = 0.900$.

Pattern 1 as v0.6 stated it replicates at v0.9. The t_2 ranking matches v0.6's narrative ordering more closely than t_1 does (p 0.900 versus 0.600), with finance cross-model spread increasing (+3.9pp) and project management spread decreasing (−1.4pp) between waves — both within H1's empirical noise floor (Finding 1) but in directions that strengthen v0.6's qualitative ordering at t_2 .

The post-hoc Pattern 1 result is reported as exploratory and is not part of the formal H1 through H6 score. It is included here because v0.6 stated Pattern 1 in qualitative terms (which categories have wider cross-model spreads) and v0.9 has the data to test that specific qualitative claim quantitatively at the longitudinal interval, even though Pattern 1 was not the variable the v0.9 pre-registration committed to. A v0.10 pre-registration could lock the Pattern 1 ordering as a formal hypothesis given the v0.9 calibration.

Limitations

Seven caveats apply to the v0.9 findings. Each is disclosed for transparency; none individually invalidates the longitudinal validity argument, but each constrains how the results generalize.

Single-pair longitudinal subset. The longitudinal claim is computed on the matched two-model subset (Anthropic Claude Sonnet 4.6 and OpenAI gpt-5.4-mini) — the only models present at both t_1 and t_2 . The four parallel-baseline models added at t_2 contribute parallel new baselines but cannot contribute to the longitudinal claim. The validity argument generalizes most directly to the matched-subset model behavior.

Seven-day inter-measurement interval. The interval is short enough that no major model release or training-cutoff change is expected to fall within it. Longer intervals (one month, three months, twelve months) are required to characterize drift across model-pipeline transitions — a v0.10+ candidate.

Gross-Presence-vs-naive-phantom-rate distinction not measured at v0.9. The H4 finding is stability of *gross Presence* — the variable v0.6 §4.6 reported and the variable on which the pre-registration tested. The v0.7 reframe distinguished naive-phantom rate (no caveat) from caveated phantom (model presents brand alongside knowledge of its decommissioning). Whether the naive-phantom rate alone is similarly stable is a separable question that a v0.10 pre-registration could address.

Registry-construction reflexivity. The K-beauty observation in Finding 5 establishes that the registry-construction process inherits Anglo-discourse coverage bias. The reflexivity does not invalidate the H1 through H6 results — which are computed on the registered brand population whatever its construction — but constrains their generalization.

Construct validity remains open. The longitudinal validity argument establishes that AI Presence behaves like a real LLM-tier construct. Whether AI Presence correlates with consumer awareness, purchase intent, or category share at the consumer-facing tier is the separate question the AIAS Phase 3 program will address. The v0.9 result unblocks Phase 3 (which requires three categories at two time points and now has data for five) but does not itself constitute construct-validity evidence at the consumer tier.

Personal finance metadata anomaly. The v0.6 personal finance measurement carries the *brand_registry_version* stamp “v2-skincare”, inherited from the prior session's

module-level constant. The metadata is cosmetic; raw-response inspection confirms that the prompts and registry that fired at v0.6 were the personal finance prompts and registry. The anomaly is disclosed for transparency.

Mode-classification audits deferred. Manual inter-rater agreement audits on the v0.9 mode-classified data have not been completed at the time of this report's posting. The v0.8 audit established 96 percent strict agreement on the brand-surfacing macro unit (the unit at which H1 through H6 score) and 68 percent strict agreement on the underlying five-mode taxonomy. The v0.9 H1 through H6 results score against the macro unit and are therefore not dependent on the strict-mode-taxonomy agreement budget. Manual audits are scheduled prior to v0.10.

What's next

The Phase 3 program will address five priorities surfaced by the v0.9 findings. Each is enumerated as a candidate v0.10 or later study with the relevant data already in place.

Construct-validity correlation. The v0.9 result unblocks the Phase 3 construct-validity study by providing five categories at two time points. Candidate consumer-tier validators include search-volume data (Google Trends), retail-tracking data (Amazon Best Seller ranks; category-share trackers where available), DTC brand-tracking instruments, and survey-based brand-awareness measurements. The construct-validity work is the program's gate to v1.0 release.

Longer-interval longitudinal measurements. Drift across one-month, three-month, and twelve-month intervals — particularly across model-pipeline transitions or training-cutoff changes — would characterize the temporal stability of AI Presence at scales that bracket model retraining, brand-discourse evolution, and consumer-behavior cycles. The v0.9 seven-day interval establishes a noise floor; longer intervals would add structural drift on top of the noise floor.

Cross-lingual category expansion. Premium tea and traditional spirits remain candidate categories for designed-for-test measurements of Pattern 4. The v0.9 H5 sensitivity — which surfaces the brand-marketing-language tier as the binding variable via the Graza diagnostic — informs the design of future cross-lingual studies: the discourse-language coverage of each registered brand becomes a primary variable, with country of origin as a secondary controlled covariate.

Naive-phantom-rate longitudinal stability. Extending the v0.7 reframe into the longitudinal frame — measuring whether Mint's naive-phantom rate is as stable as its gross Presence — would distinguish the recommendation-slot persistence component of phantom-brand surfacing from the cumulative-discourse-mass component. A v0.10 pre-registration on this variable specifically is feasible against the v0.9 dataset (raw responses are deposited; only the classifier needs to be applied).

Registry-construction protocols for cross-lingual categories. The K-beauty observation suggests a structural improvement to the AIAS registry-construction methodology. A formal protocol that explicitly probes for non-English-discourse coverage gaps (for example, through the requirement that any registry intended to measure Pattern

4 in a category include a stated population-share survey of non-English-discourse producers) would mitigate the reflexivity caveat. The protocol revision is planned as part of AIAS Presence Measurement Protocol v1.2.

Pre-registration outcomes

Six pre-registered hypotheses plus the H5 strict-vs-inclusive sensitivity check, plus an exploratory mode-distribution observation, plus a post-hoc Pattern 1 replication test. **Headline: five formal hypotheses confirmed; the sixth landed at the predicted stability band.**

H	Pre-registered prediction	Result	Status
H1	Per-brand drift small and tightly bounded (↯70% within ±5pp; ↯90% within ±10pp).	88.3% within ±5pp; 99.0% within ±10pp.	CONFIRMED
H2	Top-3 at t_1 remain in top-5 at t_2 , in 5 of 5 categories.	5 of 5; no exceptions.	CONFIRMED
H3	Within-category brand-presence variance ordering preserves (Spearman ρ ↯ 0.7).	$\rho = 0.8$.	CONFIRMED
H4	Mint matched-subset gross Presence stable within ±5pp (stability predicted, not decay).	−3.1pp (44.8% → 41.7%); within band.	STABILITY (predicted)
H5 strict	Spanish olive oil aggregate ↯12.5pp AND K-beauty aggregate ↯5pp.	11.5pp; 1.0pp — both pass.	CONFIRMED
H5 inclusive	As H5 strict, with Graza included in the Spanish cohort.	14.2pp (breach); 1.0pp.	PARTIALLY CONFIRMED
H6	Per-brand cross-model spread Pearson r ↯ 0.7 between t_1 and t_2 , in ↯4 of 5 categories.	5 of 5.	CONFIRMED
Mode dist.	Exploratory; no pre-registered threshold. Reported as a calibration baseline for v0.10.	Largest shifts: PM software +7.3pp brand mode; olive oil −8.3pp brand mode; skincare −6.2pp component mode.	EXPLORATORY
Pattern 1 (post-hoc)	Spearman ρ on cross-model spread ranking, t_1 vs t_2 ; and t_2 vs v0.6 narrative ordering.	$\rho = 0.800$ (t_1/t_2); $\rho = 0.900$ ($t_2/v0.6$).	POST-HOC REPLICATED

Per-hypothesis breakdown

Each pre-registered hypothesis with what the test measured, what the data delivered, and how to read the outcome against the construct-validity argument.

- H1 — Drift.** Per-brand magnitude of $t_1 \rightarrow t_2$ Presence delta on the matched subset. Establishes the empirical noise floor for one-week measurement intervals: **±5pp for the modal brand, ±10pp for ninety-nine of every hundred.** The absence of large-magnitude drift is the empirical signature of a stable underlying quantity.

H2 — Leaderboards. Whether top-of-category brands at t_1 remain top-of-category at t_2 . **5 of 5 categories; every t_1**

- top-three brand in the t_2 top-five.** The most directly category-management-relevant of the v0.9 findings: top-of-leaderboard composition behaves as a stable leading indicator across at least the seven-day interval.

H3 — Within-category variance. Whether categories preserve their within-category brand-presence dispersion ordering across the wave. **Spearman $\rho = 0.8$** across the 5 categories — above the 0.7 threshold for confirmation. Personal finance is

the highest-dispersion category at both waves; running shoes the lowest. The structural shape of within-category fragmentation is preserved across the longitudinal interval.

H4 — Mint persistence. Whether the v0.6 phantom-brand observation (Mint at 44 percent gross Presence despite operational decommissioning in March 2024) is stable at t_2 . **44.8% → 41.7% (−3.1pp)**; within the predicted ± 5 pp stability band. The v0.7 reframe distinguishing naive-phantom rate from gross Presence is not unpacked at v0.9; the variable v0.6 §4.6 reported is what H4 confirms longitudinal stability for.

H5 — Pattern 4. Whether the v0.6 §4.4 discourse-language bias hypothesis replicates with explicit numerical thresholds. **Strict reading: 11.5pp Spanish (below 12.5pp threshold); 1.0pp K-beauty (below 5pp threshold).** Inclusive reading with Graza added: 14.2pp Spanish (breach). The v0.9 contribution is operationalizing the v0.6 brand-marketing-language framing at threshold and surfacing Graza as the diagnostic case

isolating marketing-language from country-of-origin.

H6 — Cross-model spread. Whether per-brand Sonnet-vs-mini spread structure preserves across waves.

Pearson $r = 0.7$ in 5 of 5 categories. Independent variance-replication test at the brand $\times$ model aggregation level, separate from H3's category-aggregation test. Together H3 and H6 establish two independent variance structures that both replicate.

Pattern 1 (post-hoc). Whether v0.6's qualitative cross-model spread ordering by category replicates at v0.9. **Spearman $\rho = 0.800$ (t_1/t_2); $\rho = 0.900$ (t_2 vs v0.6 narrative ordering).** The t_2 ranking matches v0.6's stated qualitative ordering more closely than t_1 does, with finance widening (+3.9pp) and project management narrowing (−1.4pp) — both within H1's empirical noise floor but in directions that strengthen v0.6's narrative.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

Authored by

Pablo Ulpiano Gonzalez Castro

School of Visual Arts, MPS Branding Program, New York, NY (primary academic affiliation)

Third System™ (research entity; data archive and methodology venue)

Correspondence: pablou@pablou.com · pablou.com

Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.1). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

Gonzalez Castro, P. U. (2026). *AI Presence Drift: AI Presence Index v0.9 — Longitudinal re-baseline of five categories*. Third System. thirdsystem.ai/v09-longitudinal-rebaseline

Underlying datasets

v0.9 raw responses, mode-classified responses, per-brand Presence rates per category per wave, and pre-registration outcome score — deposited as the v0.9 OSF release alongside the v0.6 Cross-Category Findings dataset.

v0.6 baseline data and v0.7/v0.8 designed-for-test datasets remain available in their respective OSF deposits (unchanged).

Build pipeline (chart generator, brand-format report builder, scoring scripts) deposited as v0.9 source archive.

Methodology log: Pre-registration locked at git commit f8cebdb (tag v0.9-prereg-locked) prior to any t_2 data collection. Methodology decisions — matched two-model subset, retroactive mode classification of v0.6 data, registry freeze at v0.6 final state — locked at git commit e6428a4 prior to the pre-registration. AIAS Presence Measurement Protocol v1.1.

Inquiries: hello@thirdsystem.ai